

HR Analytics Dashboard – Business Insights Report

Project Objective

The objective of this project is to analyse employee data to understand workforce distribution, attrition patterns, employee demographics, and factors affecting employee retention. The dashboard helps HR managers make data-driven decisions related to hiring, retention, and workforce planning.

Tools Used

- Python (Pandas, NumPy, Matplotlib, Seaborn)
- SQL (MySQL)
- Power BI
- Excel

Dataset Summary

Metric	Value
Total Employees	1473
Active Employees	1236
Attrition Count	237
Attrition Rate	16%
Average Age	36.92 Years
Average Monthly Income	₹6.50K

Executive Summary

The HR Analytics dashboard provides a comprehensive overview of the organization's workforce. It highlights employee demographics, department distribution, attrition trends, salary analysis, and workforce characteristics.

The analysis indicates that employee attrition is **16%**, with the majority of employees working in the Research & Development department. Most employees belong to the **26–35 years** age group, and the largest tenure group consists of employees with **4 years** at the company. These insights can help HR identify retention opportunities and improve workforce planning.

Business Insights

1. Workforce Size

- The organization currently employs **1473 employees**.
- Out of these, **1236 employees are active**, while **237 employees have left the company**.

2. Department-wise Distribution

- Research & Development has **963 employees**.
- Sales has **447 employees**.
- Human Resources has **63 employees**.

Insight

Research & Development is the largest department, accounting for the majority of the workforce.

3. Attrition Analysis

- Overall Attrition Rate = **16%**

The dashboard indicates employee attrition across different departments.

Insight

The company should monitor departments experiencing higher employee turnover and investigate potential causes such as workload, career growth, compensation, or work-life balance.

4. Gender Distribution

- Male = **884 (60.01%)**
- Female = **589 (39.99%)**

Insight

The workforce consists of approximately **60% male** and **40% female** employees.

5. Job Role Distribution

The largest job roles include:

- Sales Executive
- Research Scientist
- Laboratory Technician

Insight

These positions form a significant portion of the workforce and should be prioritized in recruitment and retention planning.

Detailed Insights

6. Average Monthly Income

The highest average monthly incomes are observed among:

- Manager
- Research Director
- Healthcare Representative

Insight

Senior leadership and specialized roles receive higher compensation compared to operational roles.

7. Education Field Distribution

- Life Sciences = **606**
- Medical = **466**
- Marketing = **159**
- Technical Degree = **132**
- Other = **83**
- Human Resources = **27**

Insight

Most employees have educational backgrounds in Life Sciences and Medical fields, indicating the company's strong focus on technical and scientific talent.

8. Marital Status Distribution

- Married = **675**
- Single = **471**
- Divorced = **327**

Insight

Married employees form the largest employee group.

9. Employee Age Distribution

- 18–25 = **123**
- 26–35 = **607**
- 36–45 = **470**
- 46–55 = **226**
- 55+ = **47**

Insight

The organization has a predominantly young to mid-career workforce, with the highest concentration in the **26–35 years** age group.

10. Employee Tenure

From your dashboard:

- **197 employees** have completed **4 years** at the company (highest visible count).

- The majority of employees have **less than 10 years** of service.
- Very few employees have **more than 20 years** of tenure.

Insight

The workforce is largely composed of employees with low to mid-level tenure, while long-serving employees represent a much smaller proportion of the organization.

Recommendations

1. Implement targeted retention programs in departments with higher attrition.
2. Review overtime policies and employee workload to improve work-life balance.
3. Continue investing in career development for employees in their early years with the company.
4. Develop succession planning and mentorship initiatives for experienced employees.
5. Strengthen employee engagement and recognition programs to improve retention and job satisfaction.

Conclusion

This project demonstrates how Python, SQL, and Power BI can be used together to transform raw HR data into meaningful business insights. The dashboard enables HR teams to monitor workforce trends, identify attrition patterns, and support strategic decision-making through data-driven analysis.